

Sam Kay

MATHEMATICIAN · MUSICIAN · PRIVATE TUTOR

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Personal Profile

As a highly motivated and dedicated student with a deep passion for learning and personal growth, I consistently go above and beyond in my studies. I am recognised for my exceptional work ethic and efficient time management skills, which have enabled me to consistently meet deadlines and attain outstanding academic results. I possess an unwavering aspiration to succeed and am always endeavouring to elevate myself to new heights of accomplishment.

I am agreeable, enthusiastic and a clear communicator. A skill I am most proud of is the ability to convey complex ideas with a flexible vocabulary, tailoring conversations to a wide variety of audiences.

Education

Durham University

Doctor of Philosophy (PhD) in Mathematical Sciences

Durham, UK

Oct 2025 - Current

- Working towards a PhD under the supervision of Dr Peter Wyper.
- Researching structures in the solar wind, which falls under the realm of magnetohydrodynamics (MHD).

Durham University

Master of Mathematics (MMath)

Durham, UK

Sept 2021 - July 2025

- Graduated with Master's degree in mathematics with First Class Honours.
- Involved with the Mathematical Society.
- **Selected Modules:** Advanced Quantum Theory, Fluid Mechanics, General Relativity, Superstrings.

New College Pontefract

A-Levels

Pontefract, West Yorkshire

Sept 2019 - June 2021

- A* in Mathematics, A* in Further Maths, A in Chemistry, AS Music.
- Awarded with Certificate of Outstanding Effort and Performance in Further Maths.

Kettlethorpe High School

GCSEs

Wakefield, West Yorkshire

Sept 2014 - June 2019

- 11 qualifications (Levels 9-6).
- Includes Mathematics 9, Physics 8, Chemistry 8, Music 9, A* with Distinction in Further Maths.

University Projects

Chalkdust magazine & Chalkboard Ultra podcast

Mathematics outreach and communication

Oct 2023 - Current

- Writing and editing for a popular maths magazine, distributed nationally among universities.
- Hosting a podcast bringing light to all areas of mathematics, with conversations among PhD students and lecturers.

The Standard Model from a Symmetry Perspective

MMath research project

Oct 2024 - May 2025

- Researched the construction of the Standard Model of Particle Physics; studied gauge theory and spontaneous symmetry breaking.
- Deepened my understanding through reading academic papers and lecture notes from various sources.
- Produced a report of my research, alongside an academic poster and presentation, available to view on my website.

On Modelling Seismic Waves

Mathematical Modelling module

Dec 2022 - Jan 2023

- Conducted research into how we can simulate the effects of seismic waves to better understand the Earth's structure.
- A solo project, programming 2-dimensional partial differential equation solvers into Python.
- Produced a typeset essay to display my findings, receiving a 1:1 (77) in my Mathematical Modelling module.

Skills

Programming $\LaTeX$ (Overleaf/TeXShop), Python, HTML, Desmos, VisualPDE.

Soft Skills Teamwork, Problem-solving, Organisation, Data analytics, Critical thinking, Report writing, Presentation skills.

Miscellaneous Microsoft Excel, Canva, sleight of hand and card magic.

Volunteering

Stephenson College JCR

Maths subject representative

- Point of guidance contact for students regarding mathematical sciences.
- An advisory role for those needing help with exam techniques, module choices, and general academic problems.

Stephenson College

Jan 2025 - Current

Butler Jazz & Butler Soul Band

President

- Leading two bands after being the lead pianist for two years and Librarian for one year prior.
- Directing rehearsals, booking gigs, organising charity events, overseeing the executive team, managing band finances.
- Improved skills in performance, public speaking, organisation and networking with other bands.

Josephine Butler College

Sept 2023 - Nov 2024

Work Experience

Durham University

Undergraduate Tutorial Leader

- Assisting undergraduate students with their studies in the form of tutorials; guiding them through chosen problem sheet questions and being a mentor.
- Teaching assistant for Fluid Mechanics III; tutorial leader for Mathematical Methods II and Methods of Mathematical Physics II.

Durham, UK

Oct 2025 - Current

Island Lake Camp

Summer Camp Counsellor

- Mentor and counsellor for children aged 6-16 at a sleep-away summer camp in America.
- Working as a part of the STEM, Outdoor Adventure and Rockshop departments, ensuring children were safe and engaged in their activities.

Starrucca, Pennsylvania

Summers 2023, 2024 & 2025

Stephenson Central

Bartender

- Responsible for managing the bar through promotions, selling, transactions, stock control, and ensuring a smooth operation.
- Cleaned and restocked whilst customer flow was slow.

Durham, UK

Sept 2023 - June 2025

MyTutor.co.uk

Maths and Science Tutor

- Tutoring in a classroom environment via video call.
- Private tutoring with 100% return business.

Remote

Jan 2022 - June 2025

Institute of Sales Professionals

Data Research Assistant and Analyst

- Supporting the creation, deployment, and analysis of survey and research projects.
- Project of interest: a study with Cranfield University, delivering insights on the impacts of coaching within sales.

Remote

Jan 2022 - Dec 2022

Achievements

'23-'25 **3x College Scholar Award** – Stephenson College

Durham University

'22-'24 **3x Competitor** – 24 Hours of Le Maths

DU MathSoc

2020 **Regional Finals Competitor** – Senior Team Maths Challenge

UKMT

2019 **The Maureen Winderbank Trophy** – KHS Presentation Event

Kettlethorpe High

2019 **Young Award for Music** – KHS Presentation Event

Kettlethorpe High

2018/19 **Head Boy** – Leadership role

Kettlethorpe High

Interests

Fitness Regularly attend the gym and enjoy running; a trained red belt in Tae Kwon-Do.

Performing Regular at Durham Jazz Society's jazz jam nights @ Fabio's bar.

Scouting Graduated through to Explorer level, began training to become a Young Leader.

Music Worked towards ABRSM Grade 7 classical piano, played the guitar since age eight.

References available upon request.

Last updated 19 November 2025.